

Assignment - 2

1. Open Reading Frames Detection:

- Create a function that finds all the ORFs in a given DNA sequence. Return a list of ORFs along with their start and end positions.
- Modify the function to filter out ORFs that are too short (less than a specified length).

2. What is the difference between FASTA and FASTQ?

3. What should be used instead of Seq object so it can be modified?

4. Create a Python program that generates a dot plot to visualize the relationships between two DNA sequences.

5. Codon Usage Table Generator:

- Create a Python function that returns a codon usage frequency table for a given DNA sequence and visualizes it.
- Modify the function to handle multiple sequences from a FASTA file and generate a combined codon usage plot.

General Instructions:

- This is an individual based assignment, that should be submitted through Google Classroom.
- You should submit a zipped file FIRSTNAME_LASTNAME which contains your code and report.
- The report file should contain an explanation of your code, pseudocode, any resources used, and answers to any questions asked in the statement.
- OPTIONAL: You can add your stress testing code as a comment in your code. You can add your test cases and debugging efforts as well.
- Your code should be clear, understandable, and documented (comments). Follow a consistent naming convention for variables, classes, and functions.

○ The due date for the submission of this assignment is Thursday, 01/05/2025 at 11:59 pm.